

Year 10 Ext Exponents Worksheet

NO CALCULATOR

Simplify.

- Your answer should contain only positive exponents.
- Show ALL steps of working

Hint: Write numbers with powers equivalent to the denominator

ie: $(25x^4)^{\frac{1}{2}} = (5^2x^4)^{\frac{1}{2}} = (5^2)^{\frac{1}{2}}(x^4)^{\frac{1}{2}} = 5x^2$

1. $(n^4)^{\frac{3}{2}}$ 2. $(27p^6)^{\frac{5}{3}}$ 3. $(25b^6)^{-1.5}$ 4. $(64m^4)^{\frac{3}{2}}$

5. $(2m^2)(4m^{\frac{3}{2}})(4m^{-2})$ 6. $(3b^{\frac{1}{2}})(b^{\frac{4}{3}})$ 7. $(p^{\frac{3}{2}})^{-2}$

8. $\frac{2x^{-\frac{6}{4}}}{4x^{\frac{4}{3}}}$

9. $\frac{(2pm^{-1}q^0)^{-4}(2m^{-1}p^3)}{2pq^2}$

Simplify. Your answer should contain only positive exponents.

10. $\left(\frac{x^{\frac{1}{2}} y^{-2}}{y x^{\frac{-7}{4}}} \right)^4$

11. $\frac{(x^{\frac{-1}{2}} y^2)^{\frac{-5}{4}}}{x^2 y^{\frac{1}{2}}}$

Simplify

$$1. (n^4)^{\frac{3}{2}}$$

$$= n^{12/2}$$

$$= n^6$$

$$2. (27p^6)^{\frac{5}{3}}$$

$$= 27^{5/3} p^{30/3}$$

$$= (\sqrt[3]{27})^5 p^{10}$$

$$= 3^5 p^{10}$$

$$= 243 p^{10}$$

$$3. (25b^6)^{-1.5}$$

$$= (25b^6)^{-3/2}$$

$$= 25^{-3/2} b^{-18/2}$$

$$= (\sqrt{25})^3 b^{-9}$$

$$= \frac{1}{5^3 b^9}$$

$$= \frac{1}{125 b^9}$$

$$4. (64m^4)^{\frac{3}{2}}$$

$$= 64^{3/2} m^6$$

$$= (\sqrt{64})^3 m^6$$

$$= 8^3 m^6$$

$$= 512 m^6$$

Simplify. Your answer should contain only positive exponents.

$$5. (2m^2)(4m^{\frac{3}{2}})(4m^{-2})$$

$$= 32 m^{2+3/2-2}$$

$$= 32 m^{3/2}$$

$$6. (3b^{\frac{1}{2}})(b^{\frac{4}{3}})$$

$$= 3 b^{\frac{1}{2} + \frac{4}{3}}$$

$$= 3 b^{\frac{3/6 + 8/6}{1}}$$

$$= 3 b^{11/6}$$

$$7. (p^{\frac{3}{2}})^{-2}$$

$$= p^{-3}$$

$$= \frac{1}{p^3}$$

$$8. \frac{2x^{-\frac{6}{4}}}{2x^{\frac{4}{3}}}$$

$$= \frac{x^{-3/2}}{2x^{4/3}}$$

$$= \frac{x^{-3/2-4/3}}{2}$$

$$= \frac{x^{-9/6-8/6}}{2}$$

$$= \frac{x^{-17/6}}{2}$$

$$= \frac{1}{2x^{17/6}}$$

$$9. \frac{(2pm^{-1}q^0)^{-4}(2m^{-1}p^3)}{2pq^2}$$

$$= \frac{(2^{-4}p^{-4}m^4q^0)(2m^{-1}p^3)}{2pq^2}$$

$$= \frac{2^{-3}p^{-1}m^3q^0}{2pq^2}$$

$$= \frac{2^{-4}p^{-2}m^3q^{-2}}{2pq^2}$$

$$= \frac{m^3}{16p^3q^2}$$

$$10. \left(\frac{x^{\frac{1}{2}}y^{-2}}{yx^{\frac{-7}{4}}} \right)^4$$

$$= \frac{x^2y^{-8}}{y^4x^{-7}}$$

$$= \frac{x^9y^{-12}}{y^4x^{-7}}$$

$$= \frac{x^9}{y^{12}}$$

$$11. \frac{(x^{\frac{-1}{2}}y^2)^{\frac{-5}{4}}}{x^2y^{\frac{1}{2}}}$$

$$= \frac{x^{5/8}y^{-10/4}}{x^2y^{1/2}}$$

$$= \frac{x^{5/8-2}y^{-10/4-1/2}}{y}$$

$$= \frac{1}{x^{11/8}y^3}$$